

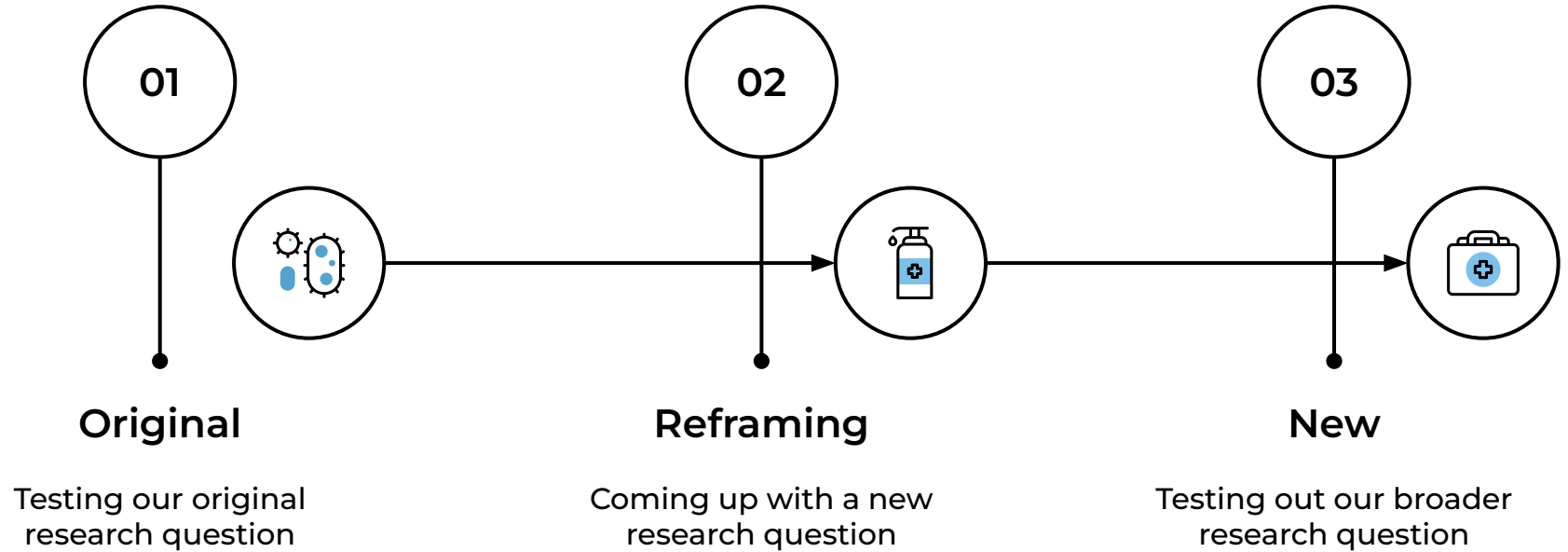


Predictive Models In ICU Databases

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DATA 4381: DATA CAPSTONE

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Updates!



Refresh!

Do predictive models for ICU **mortality** or **readmission** risk **exhibit bias** across demographic groups or hospital systems?



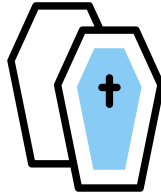
Target: Mortality - 0 if alive, 1 if died

	Precision	Recall	F1
0	0.92	1	0.96
1	0	0	0
Accuracy			0.92
Macro Avg	0.46	0.50	0.48

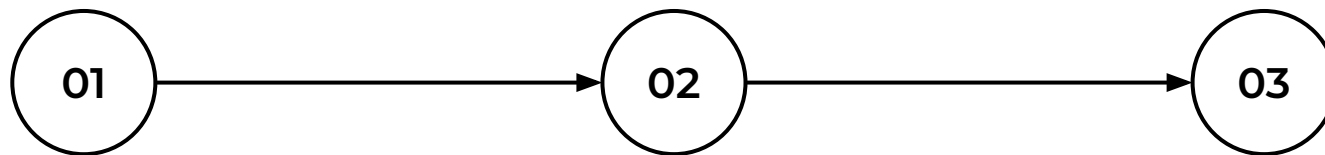
Columns		What columns to keep! Lit review time!			
Patient Demographics	age, gender, ethnicity, unittype	Columns		Columns	
APACHE Metrics	acutephysiologyscore, apachescore, apacheversion			labs 24hr	glucose_mean, creatinine_mean, bun_mean, sodium_mean, potassium_mean
Predicted Outcomes	predictedicumortality, predictedhospitalmortality			hospital context	region, managementsystem
Actual Outcomes	actualicumortality, actualhospitalmortality			provider context	physicianspeciality
Early Vital Signs	hr_mean, rr_mean, sao2_mean, temp_mean, map_mean			predicted vs actual outcomes	predictedhospitalmortality, actualhospitalmortality, predictedicumortality, actualicumortality
		Early Labs	glucose_mean, creatinine_mean, bun_mean, sodium_mean, potassium_mean		
		Discharge Context	hospitaldischargestatus, hospitaldischargelocation		
		Readmission Setup	predictedicumortality, predictedhospitalmortality		
		Underlying Conditions	diabetes, cirrhosis, aids, immunosuppression, metastaticcancer, lymphoma		
		Vitals 24hr	hr_mean, rr_mean, sao2_mean		

Merging Tables

Darlene's attempt merging in SQLite



Final Base Tables Overview



Mortality

- Patient info & some demographics
- APACHE (predictions & outcomes)
- 24-hr vitals/labs

Readmission

- Patient info & some demographics
- Discharge info
- Hospital setup
- 24-hr vitals/labs

Bias

- Demographics
- Provider
- Hospital context for fairness analysis

BIAS

Table: bias_base																		
Filter in any column																		
	patient	unit	stayid	age	gender	ethnicity	region	managementsystem		physicianspecialty		predictedhospitalmortality		actualhospitalmortality		predictedcummortality		actualcummortality
								Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	
1			141765	87	Female	Caucasian	3		1	hospitalist	3.5813944311032138E-2	ALIVE		0.01231145545549805	ALIVE			
2			143870	76	Male	Caucasian	3		1	family practice	0.0289321626120858	ALIVE		1.5706796732545089E-2	ALIVE			
3			144815	34	Female	Caucasian	3		1	internal medicine	3.6191129716436361E-3	ALIVE		2.1330459623241791E-3	ALIVE			
4			145427	61	Male	Caucasian	3		1	surgery-trauma	1.6063934736413411E-2	ALIVE		7.5555085721478212E-3	ALIVE			
5			147307	55	Female	Caucasian	3		1	unknown	3.4700522625648899E-3	ALIVE		1.9315974648549681E-3	ALIVE			
6			147784	60	Female	Hispanic	3		1	critical care medicine (CCM)	8.990514940981147E-2	ALIVE		0.04794122246322724	ALIVE			
7			148611	28	Male	Caucasian	3		1	internal medicine	2.1927357783773972E-3	ALIVE		1.27355354932637E-3	ALIVE			
8			149713	> 89	Female	Caucasian	3		1	family practice	9.0603182379846672E-2	ALIVE		3.2460274973360508E-2	ALIVE			
9			151179	59	Female	Caucasian	3		1	pulmonary/CCM	0.68519956214267197	EXPIRED		0.55737755607065542	ALIVE			
10			151867	44	Male	Caucasian	3		1	surgery-general	5.369379580605808E-3	ALIVE		2.7126127975503878E-3	ALIVE			

READMISSION

Table:	readmission_base										Filter in any column									
	age	gender	ethnicity	unittype	hospitaldischargestatus	hospitaldischargelocation	readmit	management	system	region	diabetes	cirrhosis	aids	immunosuppression	metastaticcancer	lymphoma	admitdiagnosis	electivesurgery	activetx	hr_mean
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter		Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	87	Female	Caucasian	Med-Surg ICU	Alive	Home	NULL	NULL	NALL	NULL	NULL	NULL	NALL	NULL	NULL	NALL	NULL	NULL	NULL	106.65739193
2	87	Female	Caucasian	Med-Surg ICU	Alive	Home	0	1	3	0	0	0	0	0	RHYTHATR	0	83.1909722222			
3	76	Male	Caucasian	SICU	Alive	Home	0	1	3	0	0	0	0	0	S-CAROTEND	0	45.449367088	0	0	45.449367088
4	34	Female	Caucasian	Med-Surg ICU	Alive	Other Hospital	0	1	3	0	0	0	0	0	ODOTHER	0	83.8		0	83.8
5	61	Male	Caucasian	SICU	Alive	Home	0	1	3	0	0	0	0	0	S-GIPERFOR	0	65.457983193	0	0	65.457983193
6	55	Female	Caucasian	Med-Surg ICU	Alive	Home	NULL	NULL	NALL	NULL	NULL	NALL	NULL	NULL	NALL	NULL	NULL	NULL	NULL	78.704374826
7	55	Female	Caucasian	Med-Surg ICU	Alive	Home	0	1	3	0	0	0	0	0	S-CAROTEND	1	87.085106382			
8	60	Female	Hispanic	Med-Surg ICU	Alive	Home	0	1	3	1	0	0	0	0	COMA	0	93.72079720		1	93.72079720
9	28	Male	Caucasian	Med-Surg ICU	Alive	Other	0	1	3	0	0	0	0	0	ODOTHER	0	79.4422310756			
10	34	Female	Caucasian	SICU	Alive	Other	NULL	NULL	NALL	NULL	NULL	NALL	NULL	NULL	NALL	NULL	NULL	NULL	NULL	95.4

MORTALITY

Table: mortality_base																							Filter in any column									
	patientunitstayid		age	gender	ethnicity	unittype	acutephysiologyscore		apachescore		apacheversion		predictedcomortality		actualcomortality		predictedhospitalmortality		actualhospitalmortality		hr_mean	rr_mean										
	Filter						Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter			Filter									
1		141765	87	Female	Caucasian	Med-Surg ICU		23		47	I Va		0.0123114554549805	ALIVE		3.5813944311032138E-2		ALIVE		83.190972222222	5.9375											
2		143870	76	Male	Caucasian	SICU		43		60	IVa		1.5706796732545089E-2	ALIVE		0.02893216261220858		ALIVE		45.4493670886076	67.0316455696203											
3		144815	34	Female	Caucasian	Med-Surg ICU		25		25	IVa		2.1330459622421791E-3	ALIVE		3.6191129716436361E-3		ALIVE		83.8	16.9227227272727											
4		145427	61	Male	Caucasian	SICU		26		37	IVa		7.5550585721478212E-3	ALIVE		1.6063934736413411E-2		ALIVE		65.4579831932773	8.65546216487395											
5		147307	55	Female	Caucasian	Med-Surg ICU		15		20	IVa		1.9351974648549681E-3	ALIVE		3.4700526265648899E-3		ALIVE		87.0851063829787	8.30851063829787											
6		147784	60	Female	Hispanic	Med-Surg ICU		53		64	IVa		0.0479412246322724	ALIVE		8.9905149409811147E-2		ALIVE		93.7202797202797	24.50909090909091											
7		148611	28	Male	Caucasian	Med-Surg ICU		16		16	IVa		1.273755354932637E-3	ALIVE		2.192735787373972E-3		ALIVE		79.44223710756972	22.019521912351											
8		149713	> 89	Female	Caucasian	Med-Surg ICU		33		57	IVa		3.2460274973360508E-2	ALIVE		9.0603182379846672E-2		ALIVE		72.4487179487179	31.3376068376068											
9		151179	59	Female	Caucasian	Med-Surg ICU		15		14	IVa		0.55373795607605542	ALIVE		0.68519956214267197		EXPIRED		98.2867132867133	26.8461538461538											
10		151867	44	Male	Caucasian	SICU		14		14	IVa		2.7126127975503878E-3	ALIVE		5.3693795880605808E-3		ALIVE		94.7425373134528	13.09149253731											

Table: apachepredvar_subset														
	patientunitstayid	region	teachtype	managementsystem	readmit	diabetes	cirrhosis	aids	immunosuppression	metastaticcancer	lymphoma	admitdiagnosis	electivesurgery	activetx
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	141765	3	0	1	0	0	0	0	0	0	0	RHYTHATR		0
2	143870	3	0	1	0	0	0	0	0	0	0	S-CAROTEND	0	0
3	144815	3	0	1	0	0	0	0	0	0	0	ODOTHER		0
4	145427	3	0	1	0	0	0	0	0	0	0	S-GIPERFOR	0	0
5	147307	3	0	1	0	0	0	0	0	0	0	S-CAROTEND	1	0
6	147784	3	0	1	0	1	0	0	0	0	0	COMA		1
7	148611	3	0	1	0	0	0	0	0	0	0	ODOTHER		0
8	149433	3	0	1	0	0	0	0	0	0	0			0
9	149713	3	0	1	0	0	0	0	0	0	0	AMI		0
10	151179	3	0	1	0	0	0	0	0	1	0	SEPSISCUT		0

How I Got There: Subsets

Table: patient_subset											
	patientunitstayid	patienthealthsystemstayid	age	gender	ethnicity	unittype	unitstaytype	hospitaladmitsource	unitdischargestatus	hospitaldischargestatus	hospitaldischargeolocation
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	141764	129391	87	Female	Caucasian	Med-Surg ICU	stepdown/other		Alive	Alive	Home
2	141765	129391	87	Female	Caucasian	Med-Surg ICU	admit		Alive	Alive	Home
3	143870	131022	76	Male	Caucasian	SICU	admit	Operating Room	Alive	Alive	Home
4	144815	131736	34	Female	Caucasian	Med-Surg ICU	admit	Emergency Department	Alive	Alive	Other Hospital
5	145427	132209	61	Male	Caucasian	SICU	admit	Emergency Department	Alive	Alive	Home
6	147306	133684	55	Female	Caucasian	Med-Surg ICU	stepdown/other	Operating Room	Alive	Alive	Home
7	147307	133684	55	Female	Caucasian	Med-Surg ICU	admit	Operating Room	Alive	Alive	Home
8	147784	134042	60	Female	Hispanic	Med-Surg ICU	admit		Alive	Alive	Home
9	148611	134686	28	Male	Caucasian	Med-Surg ICU	admit	Emergency Department	Alive	Alive	Other
10	149432	135332	34	Female	Caucasian	SICU	stepdown/other	Emergency Department	Alive	Alive	Other

Created subsets to keep one row per ICU stay

Table: apachepatientresult_subset													
	apachescore	apacheversion	predictedicumorlality	actualicumorlality	predictedhospitalmortality	actualhospitalmortality	predictediculos	actualiculos	predictedhospitallos	actualhospitallos	actualventdays	physicianspeciality	physicianinterventioncatge
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	47	IV	8.2471913877810981E-3	ALIVE	3.7319948856373762E-2	ALIVE	0.722231399105669	1.5625	2.88170958065951	1.8222		hospitalist	Unknown
2	47	IVa	0.01231145545459805	ALIVE	3.5813944311032138E-2	ALIVE	1.37480651018718	1.5625	3.17392534799226	1.8222		hospitalist	Unknown
3	60	IVa	1.5706796732545089E-2	ALIVE	0.02893216261220858	ALIVE	3.0216711758903	0.5506	6.03262662378308	0.8465		family practice	Unknown
4	60	IV	1.7738751589020281E-2	ALIVE	0.02820647395637314	ALIVE	3.00652209166107	0.5506	5.9552924047471	0.8465		family practice	Unknown
5	25	IV	1.8341987534881509E-3	ALIVE	4.2529613427439049E-3	ALIVE	0.592445507010167	0.7784	2.19629430867699	0.8063		internal medicine	I
6	25	IVa	2.1330459623421791E-3	ALIVE	3.6191129716436361E-3	ALIVE	0.806311151992014	0.7784	2.20761389953117	0.8063		internal medicine	I
7	37	IV	9.514384325165182E-3	ALIVE	2.1406808566348679E-2	ALIVE	3.18810856928262	0.9506	10.9306412988452	3.6618		surgery-trauma	Unknown
8	37	IVa	7.555085721478212E-3	ALIVE	1.6063934736413411E-2	ALIVE	3.50354022395167	0.9506	10.3832369083441	3.6618		surgery-trauma	Unknown
9	20	IV	1.6086431260178809E-3	ALIVE	2.513610574104010E-3	ALIVE	0.674527080624368	0.3305	0.758865344255698	0.7674		unknown	Unknown
10	20	IVa	1.9315974648459681E-3	ALIVE	3.4700522625648899E-3	ALIVE	0.820675526587168	0.3305	1.49240572834366	0.7674		unknown	Unknown

How I Got There: Built 24-hr views for labs & vitals

- Created views summarizing the first 24-hour vitals/labs in the ICU
- The views compute mean/min/max for key vitals (HR, RR, SaO₂, Temp, MAP) and core labs (glucose, Na, K, creatinine, BUN) → one summarized row per stay

	patientunitstayid	glucose_mean	glucose_min	glucose_max	sodium_mean	potassium_mean	chloride_mean	bicarbonate_mean	calcium_mean	creatinine_mean	creatinine_min	creatinine_max	bun_mean	wbc_mean	hgb_mean	t
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	141765	75.0	61	89	139.0	4.1	105.0	21.0	8.3	1.04	1.04	1.04	28.0	10.2	12.3	
2	143870	131.5	123	140	136.0	4.4	104.0	23.5	8.0	1.015	0.89	1.14	13.5	11.7	12.3	
3	144815	82.0	82	82	141.0	3.7	105.0	26.0	8.2	0.63	0.63	0.63	6.0	7.9	12.3	
4	145427	128.5	118	139	141.0	4.5	109.0	27.0	9.5	1.05	1.05	1.05	14.0	21.1	13.3	
5	147306	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	0.65	0.65	0.65	NULL	13.5	12.1	
6	147784	137.833333...	127	148	143.0	4.3	102.0	40.5	7.9	0.64	0.64	0.64	17.0	10.7	12.8	
7	148611	112.0	112	112	142.0	3.7	106.0	30.0	8.5	0.82	0.82	0.82	7.0	7.3	15.5	
8	149432	99.0	99	99	143.0	3.6	108.0	24.0	7.8	0.6	0.6	0.6	14.0	7.2	12.7	
9	149713	97.0	97	97	133.0	3.9	99.0	22.0	8.6	0.61	0.61	0.61	11.0	9.9	12.8	
10	149714	117.0	117	117	133.0	3.4	99.0	24.0	7.9	0.61	0.61	0.61	11.0	8.2	12.0	

The Big Challenges I hit (& what I learned)

- **Deduping:** learned PARTITION BY to pick the best record per ICU stay.
- **24-hr window:** kept only first-day data (0–1440 min) to stay consistent
- **Join choices:** used INNER vs LEFT joins carefully to keep all valid patients.
- **Cleaner pipeline:** subsets + views = organized, reusable SQL (not one giant query).

```
1 SELECT readmit, COUNT(*) FROM readmission_base GROUP BY 1 ORDER BY 1;
```

	readmit	COUNT(*)
1	NULL	315
2	0	2107
3	1	98

```
1 SELECT
2 SUM(hr_mean IS NOT NULL) AS with_vitals,
3 SUM(glucose_mean IS NOT NULL) AS with_labs
4 FROM mortality_base;
```

	with_vitals	with_labs
1	1806	1680

```
15 WHEN LOWER(labname) LIKE '%platelet%' THEN 'platelets'
16 WHEN LOWER(labname) LIKE '%calcium%' THEN 'calcium'
17 WHEN LOWER(labname) LIKE '%bicarbonate%' OR LOWER(labname) = 'hco3' THEN 'bicarbonate'
18 WHEN LOWER(labname) LIKE '%lactate%' THEN 'lactate'
19 WHEN LOWER(labname) = 'ph' THEN 'ph'
20 WHEN LOWER(labname) LIKE 'pco2%' OR LOWER(labname) = 'pco2' THEN 'pco2'
```

```
1 CREATE TABLE apachepatientresult_one AS
2 WITH ranked AS (
3 SELECT
4 apr.*,
5 -- rank inside each icu stay group
6 ROW_NUMBER() OVER (
7 PARTITION BY patientunitstayid
8 ORDER BY
9 -- prefer rows that have predictions filled in
10 (predictedhospitalmortality IS NOT NULL) DESC,
11 (predictedicdmortality IS NOT NULL) DESC,
12 apacheversion DESC, -- if multiple versions, take the newest one
13 apachepatientresultid DESC -- final tiebreaker: highest id
14 ) AS rn
15 FROM apachepatientresult apr
16
17 -- select only the top ranked record per stay
18 SELECT
19 patientunitstayid,
20 acutephysiologyscore,
21 apachescore,
22 apacheversion,
23 predictedicdmortality,
```

```
CREATE VIEW IF NOT EXISTS vital_24h_features AS -- summarized feature view
WITH v AS (
SELECT
patientunitstayid,
heartrate,
respiration,
sao2,
temperature,
systemicmean
FROM vitalPeriodic
WHERE observationoffset BETWEEN 0 AND 1440 -- data from the first 24 hrs icu stay
)
SELECT
patientunitstayid,
-- finding all mean/max/min of important vitals
-- heart rate
AVG(heartrate) AS hr_mean,
MIN(heartrate) AS hr_min,
MAX(heartrate) AS hr_max,
COUNT(heartrate) AS hr_count,
-- all respiratory
AVG(respiration) AS rr_mean,
MIN(respiration) AS rr_min,
MAX(respiration) AS rr_max,
COUNT(respiration) AS rr_count,
-- oxygen saturation (SaO2)
AVG(sao2) AS sao2_mean,
MIN(sao2) AS sao2_min,
MAX(sao2) AS sao2_max,
COUNT(sao2) AS sao2_count,
```

```

1 -- FINALLY new table for MORTALITY
2 CREATE TABLE mortality_base AS
3 SELECT
4     p.patientunitstayid,
5
6     -- patient info
7     p.age,
8     p.gender,
9     p.ethnicity,
10    p.unittype, -- icu type (micu, sicu, etc.)
11
12    -- APACHE stuff (predictions & outcomes)
13    apr.acutephysiologysoore,
14    apr.apachescore,
15    apr.apacheversion,
16    apr.predictedicumortality, -- predicted icu mortality
17    apr.actualicumortality, -- actual icu outcome
18    apr.predictedhospitalmortality, -- predicted hospital mortality
19    apr.actualhospitalmortality, -- actual hospital outcome
20
21    -- vitals & labs (first 24 hrs created earlier)
22    v.hr_mean, v.rr_mean, v.sao2_mean, v.temp_mean, v.map_mean,
23    l.glucose_mean, l.creatinine_mean, l.bun_mean, l.sodium_mean, l.potassium_mean
24
25 FROM patient_subset p
26 INNER JOIN apachepatientresult_one apr USING (patientunitstayid)
27 LEFT JOIN vital_24h_features v USING (patientunitstayid)
28 LEFT JOIN lab_24h_features l USING (patientunitstayid)

```

```

1 -- base READMISSION table
2 CREATE TABLE readmission_base AS
3 SELECT
4     p.patientunitstayid,
5     p.patienthealthsystemstayid, -- links multiple stays for same patient
6
7     -- patient info
8     p.age,
9     p.gender,
10    p.ethnicity,
11    p.unittype, -- ICU type (MICU, SICU, etc.)
12
13    -- discharge details
14    p.hospitaldischargestatus, -- alive/expired at discharge
15    p.hospitaldischargelocation, -- home / hospice / skilled nursing facility, etc.
16
17    -- readmission & hospital setup
18    apv.readmit, -- readmission (0/1)
19    apv.managementsystem, -- open vs closed icu system
20    apv.region, -- hospital region (categorical)
21
22    -- underlying conditions
23    apv.diabetes,
24    apv.cirrhosis,
25    apv.aids,
26    apv.immunosuppression,
27    apv.metastaticcancer,
28    apv.lymphoma,
29
30    -- additional context cols
31    apv.admitdiagnosis,
32    apv.electivesurgery,
33    apv.activetz,
34
35    -- vitals (24hr)
36    v.hr_mean, -- heart rate
37    v.rr_mean, -- respiratory rate

```

```

1 -- base BIAS table
2 CREATE TABLE bias_base AS
3 SELECT
4     p.patientunitstayid,
5
6     -- demographics
7     p.age,
8     p.gender,
9     p.ethnicity,
10
11    -- hospital context
12    apv.region,
13    apv.managementsystem,
14
15    -- provider
16    apr.physicianspecialty,
17
18    -- predictions vs outcomes
19    apr.predictedhospitalmortality,
20    apr.actualhospitalmortality,
21    apr.predictedicumortality,
22    apr.actualicumortality
23
24 FROM patient_subset p
25 INNER JOIN apachepatientresult_one apr USING (patientunitstayid)
26 LEFT JOIN apachepredvar_subset apv USING (patientunitstayid);

```

Mortality Merged Model

Data Cleaning & Preprocessing

- **Fixed age format** (e.g., ">89" → 90)
- **Made binary mortality label** (Expired = 1)
- **Removed impossible vitals** (<10 MAP, 0 HR, SaO₂ < 50 %)
- **Imputed missing labs/vitals by ICU type** (median)
- **Normalized skewed labs** (log + winsorization)

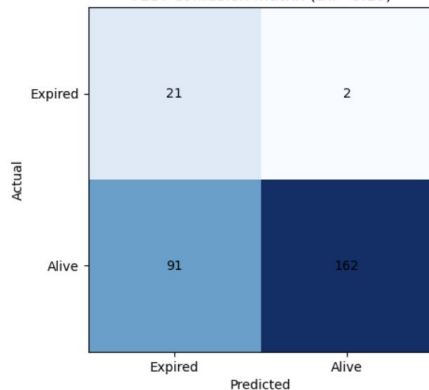
Model Setup

- **Model:** Random Forest Classifier
- **Target:** actualmortality (Expired vs Alive)
- **Features:** demographics, APACHE scores, 24-hr labs & vitals
- **Split:** 70/15/15 (train/val/test, stratified)
- **Metrics:** Precision, Recall, F1, AUROC
- Tuned probability threshold on validation

Threshold Tuning & Results

TP=21 FN=2 FP=91 TN=162

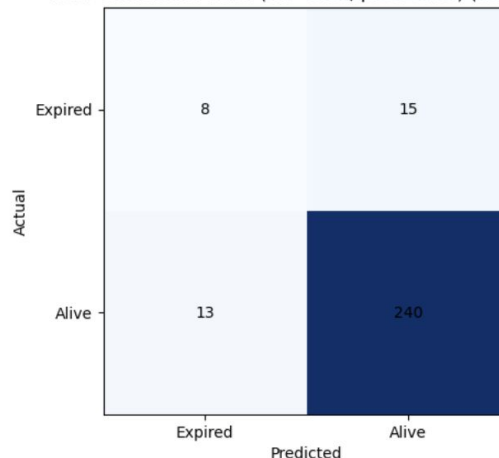
TEST confusion matrix (thr=0.10)



```
23]: recall = 21 / (21 + 2)
precision = 21 / (21 + 91)
f1 = 2 * (precision * recall) / (precision + recall)
print(f'Recall={recall:.3f}, Precision={precision:.3f}, F1={f1:.3f}')
Recall=0.913, Precision=0.188, F1=0.311
```

Chosen VAL threshold (precision ≥ 0.25): 0.30 Recall=0.125 Precision=0.250 F1=0.167 AUROC=0.755
[TEST@best_t (prec ≥ 0.25)] thresh=0.30 Recall=0.348 Precision=0.381 F1=0.364 AUROC=0.855
TP=8 FN=15 FP=13 TN=240

TEST Confusion Matrix (thr=0.30, prec ≥ 0.25) (thr=0.30)



- $t = 0.10$: Recall = 0.91, Precision = 0.19 $\rightarrow$ high sensitivity, many false alarms
- $t = 0.30$: Recall = 0.35, Precision = 0.38, AUROC ≈ 0.86 $\rightarrow$ more balanced
- Shows recall-precision trade-off

Takeaways & Next Steps

Key Takeaways:

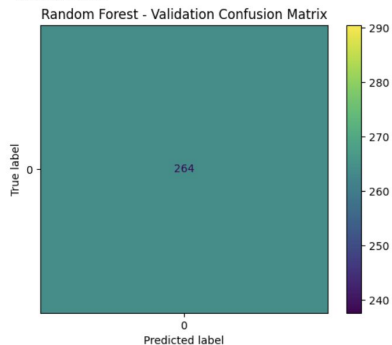
- Random Forest achieved decent AUROC (~0.86) but **still affected by class imbalance**
- Lower thresholds **improved recall but reduced precision**, highlights trade-off
- Cleaning and feature scaling steps worked, but **more EDA needed on distributions**

Next Steps:

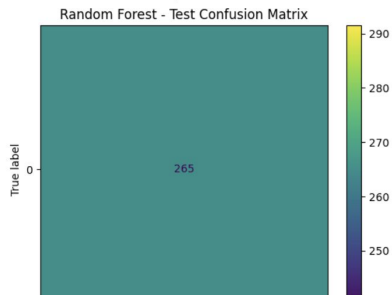
- **Redo EDA & Preprocessing** to check for outliers, skew, and feature correlations
- **Feature Importance Analysis** to identify which vitals/labs influence mortality most
- **Hyperparameter Tuning** (n_estimators, max_depth, class_weight)
- **Try Logistic Regression & XGBoost for comparison**
- **Evaluate Subgroup Bias** (gender, ethnicity, ICU type)

Random Forest and Overfitting

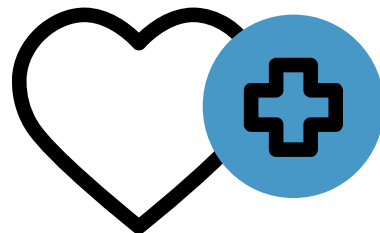
```
Training set: 1232 samples
Validation set: 264 samples
Test set: 265 samples
Random Forest Validation Accuracy: 1.0000
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1228: UserWarning:
  warnings.warn()
```



```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1228: UserWarning:
  warnings.warn()
Random Forest Test Accuracy: 1.0000
```



- Encoded target variable (Actual ICU Mortality)
- Encoded categorical variables
- Converted float values into integers
- Model wasn't picking up minority class??



We realized the real lies with our real eyes



Maybe an unbalanced dataset isn't a problem...

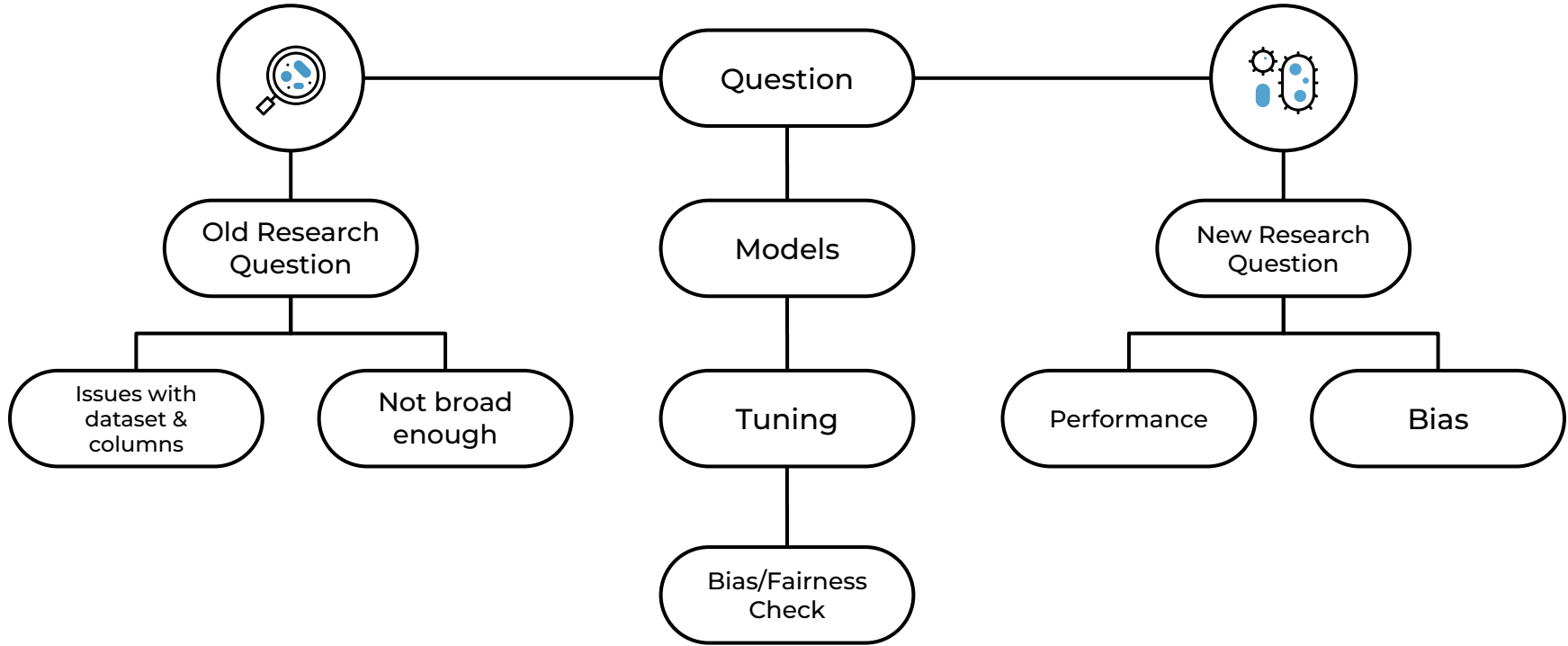
Why it may be

- Bias towards majority class
- Evaluation can be misleading
- Accuracy is not the best metric to use

Why it may not be

- Reflects real world frequencies
- Some metrics are robust to imbalance
- Balancing can distort interpretation

Where do we go from here?



Reframing

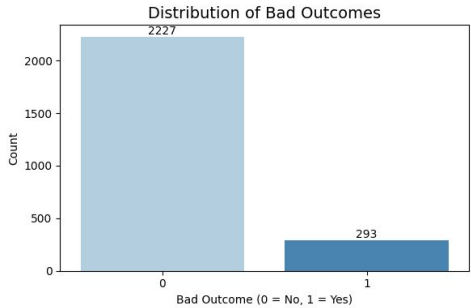
Old Question	New Research Question
What patterns of performance disparity or bias are present in ICU mortality and readmission models when comparing patients of different demographics or hospital contexts, and how might these nuances influence clinical decision-making?	Can we predict ICU patient poor outcomes (death or readmission) using demographic, clinical, and physiological data from the dataset, and how does imbalance affect model performance?



New data frame created

Darlene's Dataset	Mine
2 days	7 hours
3 target variables (readmission, hospital death, ICU death)	Merged the minority group into bad_outcomes ~ 293:2227
Created subsets by ranking	Aggregated features using mean or max
Essential vitals only, focused on details	Added comorbidities and intake/output
Kept prediction variables (i.e. predictedHospitalMortality)	Took out prediction variables to prevent data leakage
Min/Mean/Max first 24 hour window labs/vitals	24 hour vital average, stdev
2520 rows, 57 columns	2520 rows, 60 columns

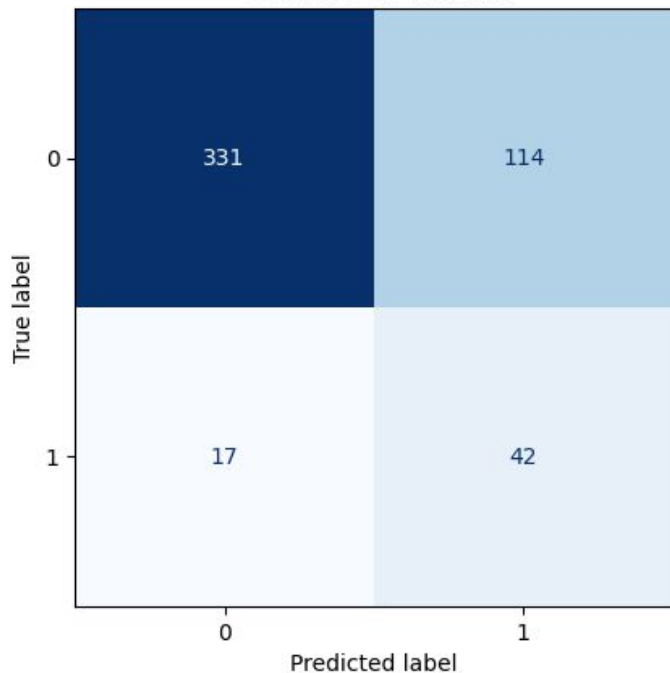
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	
	patientid	gender	age	ethnicity	hospitalid	unittype	admissionheight	admissionweight	dischargeweight	unitadmission	hospitaladmission	numdischarge	category	teachingstatus	region	acutephysiologyscore	apchscore
1																	
2	141764	Female	87	Caucasian	59	Med-Surg ICU	157.5			ICU to SDU		<100	f	Midwest			
3	141766	Female	87	Caucasian	59	Med-Surg ICU	157.5	46.5	45	Emergency Department		<100	f	Midwest	23	47	
4	143870	Male	76	Caucasian	68	SICU	187	77.5	79.4	Operating Room	Operating Room	<100	f	Midwest	43	60	
5	144816	Female	34	Caucasian	56	Med-Surg ICU	172.7	60.3	60.7	Emergency Departm	Emergency Departm	<100	f	Midwest	26	35	
6	145437	Male	81	Caucasian	68	SICU	177.8	91.7	93.1	Operating Room	Emergency Departm	<100	f	Midwest	26	37	
7	147396	Female	55	Caucasian	63	Med-Surg ICU	157.5			ICU to SDU	Operating Room	100 - 249	f	Midwest			
8	147397	Female	55	Caucasian	63	Med-Surg ICU	157.5	72.5	72.5	Operating Room	Operating Room	100 - 249	f	Midwest	15	20	
9	147784	Female	60	Hispanic	67	Med-Surg ICU	154.9	90.6	97.6	Emergency Department		<100	f	Midwest	53	64	
10	148611	Male	28	Caucasian	61	Med-Surg ICU	182.9	161.8	91.9	Emergency Departm	Emergency Departm	<100	f	Midwest	16	16	
11	149432	Female	34	Caucasian	68	SICU	165.1			ICU to SDU	Emergency Departm	<100	f	Midwest			
12	149433	Female	34	Caucasian	68	SICU	165.1	60.7	60.8	Emergency Departm	Emergency Departm	<100	f	Midwest			
13	149713	Female	> 89	Caucasian	60	Med-Surg ICU	157.5	58.5	58.6	Emergency Departm	Emergency Departm	<100	f	Midwest	33	57	
14	149714	Female	> 89	Caucasian	60	Med-Surg ICU	157.5			ICU to SDU	Emergency Departm	<100	f	Midwest			
15	151179	Female	59	Caucasian	66	Med-Surg ICU	149.9			73	Emergency Departm	Emergency Departm	100 - 249	f	Midwest	125	141
16	151867	Male	44	Caucasian	68	SICU	172.7			84.1	Operating Room	Operating Room	<100	f	Midwest	14	14
17	151900	Female	66	Caucasian	73	MICU	165.1	86.8	79	Emergency Departm	Emergency Departm	>= 500	f	Midwest	45	68	
18	152954	Female	41	Caucasian	71	Med-Surg ICU	170.2	81	80.6	Emergency Departm	Emergency Departm	100 - 249	f	Midwest			
19	153972	Male	63	Caucasian	59	Med-Surg ICU	172.7			101.4	Emergency Departm	Emergency Departm	<100	f	Midwest	24	35
20	155961	Female	57	Caucasian	56	Med-Surg ICU	157.5	120.1	123.4	Operating Room	Operating Room	<100	f	Midwest	26	31	
21	156308	Male	87	Caucasian	60	Med-Surg ICU	172.7	86.18		Emergency Departm	Emergency Departm	<100	f	Midwest	41	75	
22	156966	Female	82	Caucasian	58	Med-Surg ICU	160	115.2	115.6	Emergency Departm	Emergency Departm	100 - 249	f	Midwest	13	18	
23	157016	Female	23	Caucasian	63	Med-Surg ICU	182.6	93.5	83.6	Floor	Floor	100 - 249	f	Midwest	29	29	
24	157427	Male	73	Caucasian	56	Med-Surg ICU	180.3	89.2									
25	158056	Male	39	Caucasian	60	Med-Surg ICU	183	163.8									
26	158057	Male	39	Caucasian	60	Med-Surg ICU	183										
27	158905	Female	87	Caucasian	63	Med-Surg ICU											
28	158906	Female	87	Caucasian	63	Med-Surg ICU											
29	159411	Female	20	Caucasian	61	Med-Surg ICU	162.6	49.2									
30	160312	Male	29	Caucasian	60	Med-Surg ICU	188										
31	160510	Male	39	Caucasian	60	Med-Surg ICU	188	75									
32	160760	Male	71	Caucasian	60	Med-Surg ICU	170.2	111.1									
33	161763	Female	18	Caucasian	58	Med-Surg ICU	162.6	55.8									
34	162502	Male	83	Caucasian	69	Med-Surg ICU	180.3	104.3									
35	162841	Female	84	Caucasian	59	Med-Surg ICU	160	78.8									
36	162842	Female	84	Caucasian	59	Med-Surg ICU	160	63.5									
37	163136	Female	47	Caucasian	73	Med-Surg ICU	177.8										
38	163137	Female	47	Caucasian	73	Med-Surg ICU	177.8	96.9									
39	163524	Female	38	Caucasian	69	Med-Surg ICU	162.6										
40	165173	Male	61	Caucasian	68	SICU	185.4	106.5									
41					69	Med-Surg ICU											



	AN	AQ	AP	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC		
1	cor	leukemia	immunosuppression	cataracts	diabetes	elective surgery	active	ventilator	net total	sa2_mean	sa2_min	heart_rate_mean	heart_rate_max	heart_rate_sd	respiration_mean	respiration_max	bad_outcome
2	0	0	0	0	0	0	0	0	-500	96.8069566	96	80.6521739	138	11.33944973			0
3	0	0	0	0	0	0	0	0	3025	96.6089565	95	83.1997222	116	8.031839196	24.7620867	39	0
4	0	0	0	0	0	0	0	0	4201	96.28481013	80	45.44836709	95	2.984819602	67.03164507	86	0
5	0	0	0	0	0	0	0	0	4500	98.63088182	96	83.8	118	10.48359965	16.92727273	42	0
6	0	0	0	0	0	0	0	0	11283.16	95.76190476	89	65.45798319	85	7.86249745	16.34920635	41	0
7	0	0	0	0	0	1	0	0	-1240	95.44117647	93	78.70434783	99	8.798918273			0
8	0	0	0	0	0	0	0	0	-665.88	94.69090909	92	87.08510638	115	7.065896335	20.02564103	33	0
9	0	0	0	0	0	1	0	1	5758.02	95.86170213	90	93.72027972	109	6.3190067	24.59090909	38	0
10	0	0	0	0	0	0	0	0	4595	95.81481481	93	81.40740741	111	15.2978858	22.43333333	38	0
11	0	0	0	0	0	0	0	0	1899	98.34782609	96	95.4	106	4.816637832			0
12	0	0	0	0	0	0	0	0	0	98	98	98	98				0
13	0	0	0	0	0	0	0	0	885	93.09917355	88	72.44871795	94	5.280901138	31.60775862	54	0
14	0	0	0	0	0	0	0	0	454	93.05714286	81	66.02692963	82	5.440232574	30.97761194	46	0
15	1	0	0	0	0	0	0	0	21286.61	96.02447052	92	98.9787324	171	13.05247269	27.32284342	48	1
16	0	0	0	0	0	1	0	0	42348.48	96.11569094	87	94.74253731	100	1.93205192	23.4	34	0
17	0	0	1	1	0	0	0	0	18467.48	97.88194444	91	84.94791667	118	8.352474388	16.02666667	22	0
18	0	0	0	0	0	0	0	0	-112	95.66197183	78	89.63723294	113	8.250128905	29.41134792	61	0
19	0	0	0	0	0	0	0	0	3281	95.93023256	81	79.01937884	90	3.784518484	15.61111111	24	0
20	0	0	0	0	0	0	1	1	1150.79	95.29078014	83	81.56944444	108	6.256895456	17.33333333	36	0
21	0	0	1	0	0	1	1	1	1049	95.7385109	84	87.15734266	98	6.248064846	22.05944056	45	0
22	0	0	0	0	0	0	0	0	-2255	93.91578947	89	84.0563838	100	5.937195143	21.8547486	33	0
23	0	0	0	0	0	0	0	0	4115.8	97.76923077	97	87.20557491	130	13.28110165	26.46133846	45	0
24	0	0	0	0	0	1	1	0	-2212	96.60649819	92	95.52631579	108	14.18403345			0
25	0	0	0	0	0	0	0	0	8853.2	95.72831579	92	75.98536585	104	13.12496988			0
26	0	0	0	0	0	0	0	0	19167.33		94	73.65714286	94	6.953386213			0
27	0	0	0	0	0	0	0	0	0								0
28	0	0	0	0	0	0	0	0	1105	96.29411765	92	71.96884111	94	7.454457945	27.47389421	45	0
29	0	0	0	0	0	0	0	0	460	97.19743223	95	67.94444444	94	8.59166885	17.91975309	42	0
30	0	0	0	0	0	0	0	0	720	95.5	93	73.70909091	96	4.499681994			0
31	0	0	0	0	0	0	0	0	360	99	95	71.2	88	5.79655096			0
32	0	0	0	0	0	0	0	0	-1155	94.16866667	82	75.14876033	88	4.742604913			0
33	0	0	0	0	0	0	0	0	1298	96.41708543	94	70.68181818	104	9.77780687			0
34	0	0	0	1	0	0	0	0	3785.88	91.37412587	81	90.57594937	110	5.521437935	24.03921569	45	0
35	0	0	0	0	0	0	0	0	43635.35	97.76	92	70.2769974	94	12.49119626	20.86666667	25	1
36	0	0	0	0	0	0	0	0	28100.23	93.510493	87	76.43877097	98	6.047890005	25.1206774	35	0
37	0	0	0	0	0	0	0	0	900	96.65517241	81	119.052092	134	4.999746838	21.08278705	41	0
38	0	0	0	0	0	0	0	0	5381	94.3597061	82	112.7611312	132	5.66064338	16.30516432	41	0
39	0	0	0	0	0	0	0	0	-1607	97.99209099	95	81.08878505	102	4.548803219	22.42790988	58	0
40	0	0	0	0	0	0	0	0	645	96.425	94	79.85017422	109	26.60325993	24.4235	30	0
41	0	0	0	0	0	0	0	0	-400								0

Rough Model Results (Logistic Regression)

Confusion Matrix

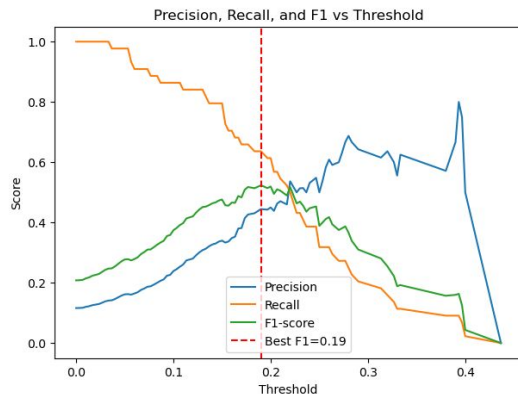


	precision	recall	f1-score	support
0	0.9511	0.7438	0.8348	445
1	0.2692	0.7119	0.3907	59
accuracy			0.7401	504
macro avg	0.6102	0.7278	0.6128	504
weighted avg	0.8713	0.7401	0.7828	504

ROC-AUC: 0.7927632831841555

- ROC-AUC: 0.79
 - Can distinguish well between good and bad outcomes
- For class 1, high recall and low precision
 - Catches most true positives but produces more false alarms

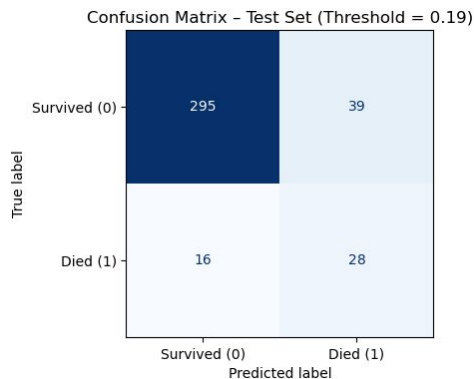
Fine-Tuning (Threshold = 0.19)



test set evaluation (w/ threshold = 0.19)

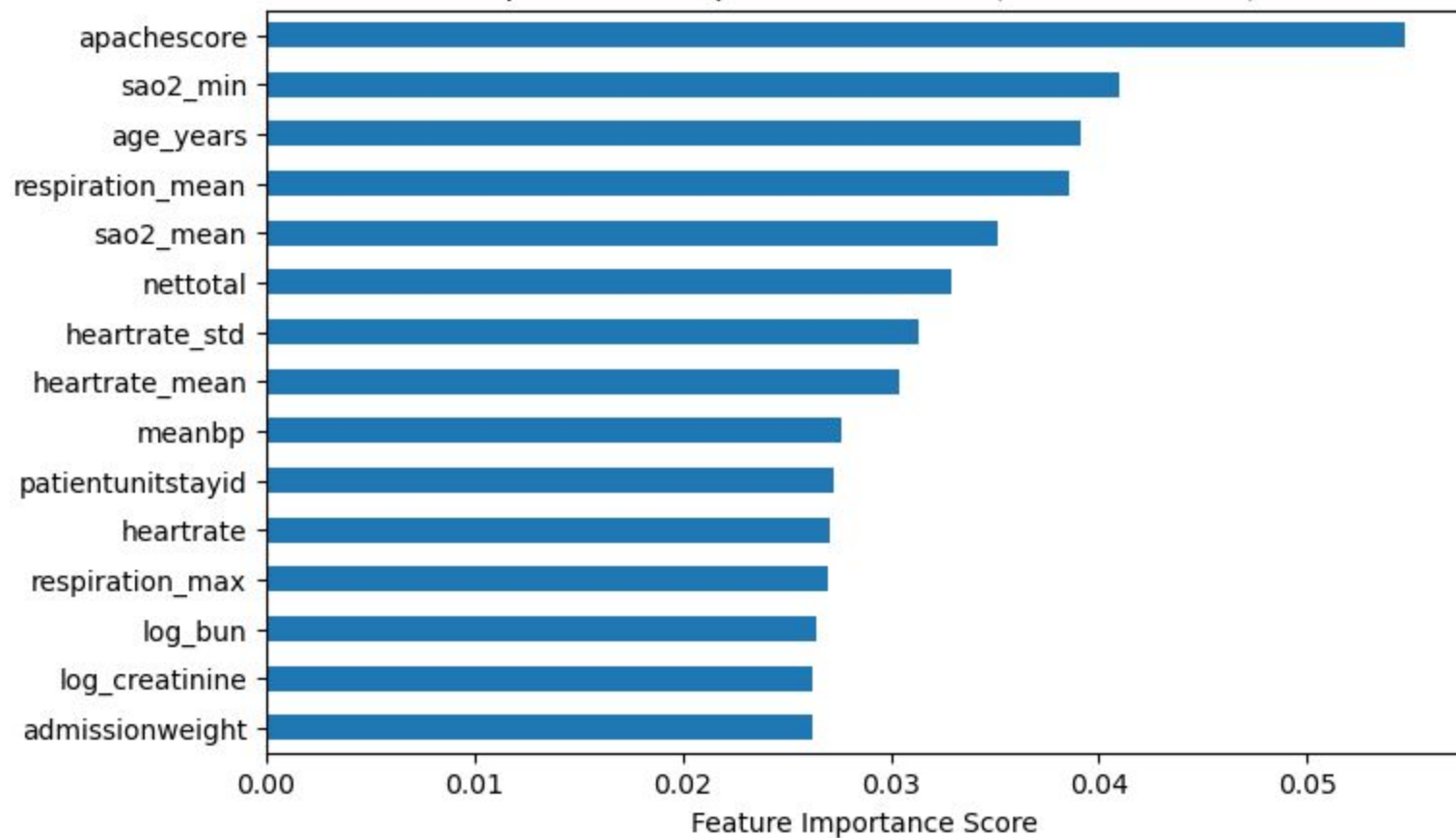
	precision	recall	f1-score	support
0	0.949	0.883	0.915	334
1	0.418	0.636	0.505	44
accuracy			0.854	378
macro avg	0.683	0.760	0.710	378
weighted avg	0.887	0.854	0.867	378

Test ROC-AUC : 0.836



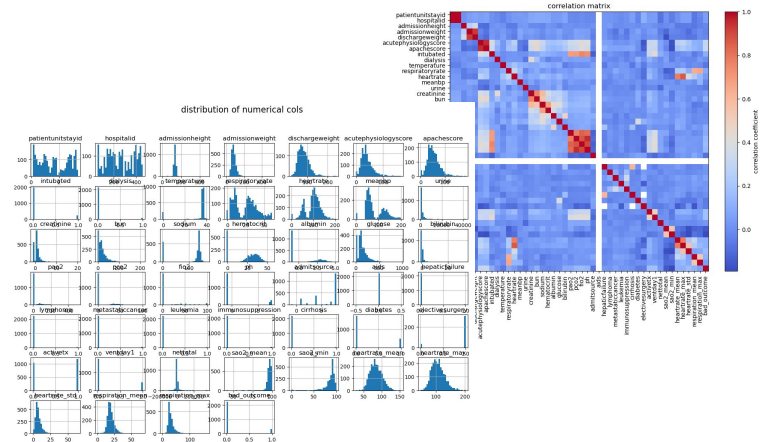
- Detected 28 out of 44 actual deaths
 - Missed 16 deaths so there are still some false negatives
- 39 false positives means model flags survivors as high risk
- ROC-AUC 0.84: model can strongly discriminate between outcomes

Top 15 Most Important Features (Random Forest)



What we'll work on now

- ❑ Officially choose which merged dataset to go with
- ❑ Figure out what to do with our null values (probably implement an “Unknown” variable)
- ❑ Do a thorough analysis (EDA) and proper cleaning of our data now that we have a data frame created
 - ❑ How to handle outliers
 - ❑ How to handle multicollinearity
- ❑ Explore fine tuning
- ❑ Look forward to fairness checks



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Thank You!

